

Figure 3.22 Profiling CPU usage when adding ordered sequences to a BST and randomized treap.

All considered these results would suggest that, if we are not sure about how uniform and duplicates free the data we have to hold are, we should consider using randomized treaps; if, instead, we are sure the data will have lots of duplicates or possibly be close to sorted, then we definitely want to avoid using plain BSTs, and resort to a balanced tree.

3.5.4 Profiling Memory Usage

So much for CPU usage, you might say, what about memory usage? Because maybe random treaps are faster in some situations, but they require so much space that you won’t be able to store them in memory for large datasets.

First, we make a consideration: memory usage will approximately be the same for all the test cases we have introduced in the previous sections (when comparing containers of the same size, of course). This is because the number of nodes in both trees won’t change with their height, these trees do not support compression and balanced and skewed trees will always need n nodes to store n keys.